

Vinith Ramaswamy

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SUMMARY

Software engineer and entrepreneur. Graduated from UW–Madison in May 2026 with Distinction, finishing a double major in Computer Science and Data Science in three years with a 3.96 GPA and honors coursework in algorithms (CS 577). Founded JustBook in second year and built it into a booking and payments platform that is live on web, iOS and Android, owning every layer from the database to the mobile apps. Now going deeper on distributed systems through Flagdown, a ride-sharing app on Go microservices, and the systems coursework behind it.

EDUCATION

B.S. Computer Science & B.S. Data Science GPA 3.96 / 4.0

Aug 2023 – May 2026

University of Wisconsin–Madison · *Graduated with Distinction, in three years* · *Certificate in Entrepreneurship*

Coursework: Big Data Systems, Honors Algorithms, Databases, Machine Learning · full list: vramaswamy4.github.io/coursework

TECHNICAL SKILLS

Languages: Python, Go, SQL, JavaScript, TypeScript, Java, C++, C, R

Backend & data: Flask, Django, SQLAlchemy, PostgreSQL, REST API design, gRPC, Redis, MongoDB, Kafka, RabbitMQ

Infrastructure: Docker, Kubernetes, Terraform, GitHub Actions, Prometheus, Grafana, OpenTelemetry, Google Cloud Run

Frontend & mobile: React 19, Next.js, Vite, Capacitor (iOS/Android), React Native, Figma

Testing: pytest, Vitest, Playwright, k6 load testing

Full list, with the project each was used in: vramaswamy4.github.io/about/#skills

EXPERIENCE

Founder & Software Engineer

Johns Creek, GA · May 2025 – Present

JustBook · *Booking and payments SaaS* · justbookapp.com · [App Store](#) · [Google Play](#) · [Write-up](#) · [Engine repo](#)

Stack: Python, Flask, SQLAlchemy, PostgreSQL, React 19, Capacitor, Stripe, Twilio, Render, AWS S3, pytest, Playwright

- **Took a booking and payments platform from idea to production in 15 months:** a Flask/PostgreSQL API (633 endpoints, 127 models), a React 19 front end and Capacitor mobile apps, now live on web, iOS and Android.
- **Designed one availability engine for every booking surface** (public site, admin calendar, waitlist) so they can never disagree about a free slot; a constant 5 queries per month view, pinned by tests; published on GitHub.
- **Eliminated double bookings under concurrent load:** reproduced the race with a k6 load test, then serialized all five booking write paths with PostgreSQL advisory locks keyed per branch and day.
- **Built the test infrastructure behind 8,500+ automated tests** (pytest on real PostgreSQL with an isolated database per run, Vitest, Playwright at desktop, phone and iOS-WebKit sizes with axe checks), so every release ships from green.
- **Replaced review-time policing with 22 custom build guards** that fail the build on secrets in logs, schema/migration drift, error-contract violations and stale documentation, so a standard cannot be broken silently.

IT Help Desk Technician, Level 1 to Lead

Madison, WI · Jan 2024 – May 2026

UW–Madison Libraries · *Part-time*

- **Promoted through four levels to Lead within a year**, one of eight leads running the public labs and help desk for 50+ student technicians on 100+ tickets a day, at 29 hours a week while a full-time student.

PROJECTS

Flagdown Full-stack ride-sharing app in Go microservices and Next.js

2026

github.com/vramaswamy4/flagdown · [Write-up](#)

Stack: Go, gRPC, RabbitMQ, Kafka, Redis, MongoDB, Kubernetes, Prometheus, Grafana, OpenTelemetry, Next.js

- **Built five Go microservices and a Next.js front end on Kubernetes**, with gRPC where the caller needs an answer, RabbitMQ where it only needs to announce an event, and WebSockets for live rider and driver updates.
- **Made the driver service stateless** by streaming positions through Kafka (keyed by driver ID) into an idempotent consumer that keeps a Redis geo index, so nearest-driver matching is one GEOSEARCH on any replica.
- **Made every ride traceable end to end:** Prometheus metrics and a provisioned Grafana dashboard on every service, with OpenTelemetry context carried through gRPC metadata and RabbitMQ headers.
- **Made message handling crash-safe:** durable queues, acks after the work, backoff retries, dead-lettering and idempotent handlers, verified by a chaos test that kills a pod mid-ride and asserts no double-processing.

APelevator AP tutoring marketplace, built 2022, rebuilt 2026 · github.com/vramaswamy4/apelevator

2022 · 2026

- **Recovered and rebuilt my 2022 high-school Django app:** 217 tests, a database-enforced token ledger, server-verified PayPal payments, an LLM study planner with a 24-case eval suite, and Terraform-managed Cloud Run with keyless CI/CD.